

Thesis “How To”: Starting your thesis work

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Slides available

<https://magrilu.github.io/tht/>

thesis how to

materials and tutorial for perspective thesis students

On this page you can find useful materials to better set up your thesis work. The content is also curated by [Prof. Boracchi](#) and it is also based on the hints and advices of [Prof. Florian](#) who is kindly acknowledged here.

The right mindset

- It's normal not to have everything clear from the start. Uncertainty is part of research, it's how discovery begins.
- The thesis journey is like moving through a jungle. You clear the path step by step. There's no single route, everyone finds their own way.
- You are not alone! Trust your advisors, they're here to support you.

Slides

1. [Getting started](#)
2. [Related Work & Problem formulation](#)
3. [Thesis preparation](#)
4. [Thesis presentation](#)

Calendar

Oggi	<	>	Set 2025 – set 2026	📅
30 SET, MAR				
17:00			THT: Thesis kick off Aula Beta Edificio 24	
1 OTT, MER				
17:00			[THT]: Thesis the final presentation 1A	
9 OTT, GIO				
15:00			THT: related work & problem formulation Aula Beta Edificio 24	



The Champion

You are the champion of your research

What we want you to:

- **become expert of your problem...**
- **propose/design** (possibly together with us) a smart solution.

We don't want you to:

- Exclusively implement some specific code / experiment...
- Exclusively do what you are instructed to do...
- Exclusively do what you want...

You are the champion of your research

- Be enthusiastic
- Be proactive
- Be curious
- Be dependable
- Ask (yourself) many questions (ask the unanswered ones to us)

... and brainstorm a lot with your advisor co-advisor!

Your Attitude

(credits Daniel Florian)

Simple attitudes toward work that can make the difference (and people happy, including you):

Be interested: *If you are not interested in what you are doing, why would your advisor be interested in helping you? Be aware that low interest or passion is hard to hide, and it is easy to guess how excited a student is about his work.*

Simple attitudes toward work that can make the difference (and people happy, including you):

Take notes: One of the simplest instruments ***to show (and stimulate) interest***, for example during a meeting, is to take notes. Bring your laptop with you or - even better - a plain old paper notebook and a pen and write down what is being said and action points, especially if they are yours. It's simply a pity if what is being said is lost immediately after the meeting and no progress happens till the next meeting.

Simple attitudes toward work that can make the difference (and people happy, including you):

Be serious: Don't be lousy or sloppy, neither with the notes you take, nor with whatever you hand over to your advisor. ***Sloppy, superficially done work leaves a very bad impression***, not to mention that it might backfire: if something is not well done, the risk is high that we won't understand. That's a failure.

Simple attitudes toward work that can make the difference (and people happy, including you):

Work hard: *Doing things right and putting the right care into them takes time. There is not much we can do about it (at least I did not yet find the magic recipe). This means you need to work hard to obtain good results and to present them in such a way that you are also proud of them. Being proud of what you produce is good and desirable.*

Simple attitudes toward work that can make the difference (and people happy, including you):

Be honest: One key ingredient to be able to be proud of your work is not to cheat or trick. If your advisor gets only the smallest feeling of dishonesty, trust is over and very likely so is your collaboration. This is a very sensible issue.

LLMs' policy:

- **Very good for searches** (but not the only source)
- **Do not write us emails using LLMs:** take time and use your own words.
- **Do not write anything you want us to check by an LLM:** asking us to review what you did not write is not honest

Add on...

Be very crical in assessing and planning your work... but do not get depressed 😊

It's better if you filter out poor/wrong ideas yourself soon

The Meetings

The meetings: very nice moments 😊

Unfortunately, we have very limited time:

- Take the most out of our meetings....
- Take notes.
- Agree on a summary at the end of the meeting.
- At the end, always define action items, todo list!

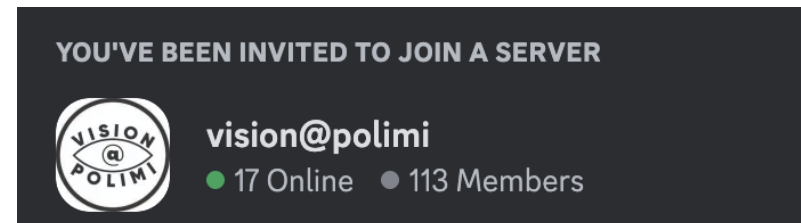
Meeting Preparation

- Need to send a calendar invite all the people involved
- Come prepared, come with
 - illustrative and clear graphics of your findings,
 - Schemes and pseudo-code of your algorithm
 - a well organized **presentation** (ppt)
- Be ready to use the whiteboard to illustrate what anything you have been doing/studying.
- No need to prepare super fancy and cool slides, don't use LLMs for slide presentation as these are typically inconsistent and not editable
- Discussion: you might have to open results for further investigation (e.g. zoom in a plot, add a curve, modify some visualization, correct something...). All your work needs to be prepared to be ready-to-use.

Interaction with the Advisor(s)

Good to know...

- Be positive... don't complain too much
- Be prepared to talk about your research at any time 😊
- Find a solution you are interested in!
- Write **email updates every Friday** (not long emails, short ones). No LLMs, learn how to craft your emails wisely.
 - It has to be clear what you are referring to
- Always include the right addresses (to: and cc:)
- Send reminders to us, if needed.
- Establish a communication channel with your co-advisor (Teams/discord). **To me Friday emails.**
- Join discord <https://discord.gg/N7dQnz652s>



Colleagues

... if you had the chance

Collaborate, discuss with others working in similar domain

We aspire at organizing presentation of our ongoing / submitted / published work

We will let you collaborate for preparing the final presentation.

Resources: <https://magrilu.github.io/tht/>

★ You can visit our [Thesis-Hall-of-Fame](#) to get an idea of our past theses (some of which have been published at workshops, conferences and journals)!

 Please fill in this [form](#) to request access to our machines with GPU and to the Airlab.

 Have a look at the guide [Servers how to](#) to setup the connection to our machines (it's a work in progress, thanks to Luca Alessandrini!).

 You can join our [Discord server](#).



Ms forms is also on the
thesis-how-to webpage



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The guide "Servers-how-to" is
available and will be updated

Access to the Lab

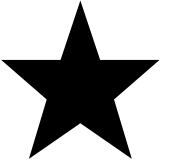


You can have access to the Airlab in Building 21 first floor.

You can request access by filling out a form.

- Convenient for you: a dedicated place to work.
- Convenient for us: more chances to meet and have quick, on-the-fly discussions, which we really value!

Best Thesis



Best thesis are those leading to

- Some advances towards the solution of a significant problem
- Some result that is worth describing in a paper

Our goal is to bring you all there. However, we need a special effort on your end.

<https://magrilu.github.io/thof/>

Thesis Hall of Fame

Here you can find some particularly worthy theses, some of which have been then evolved into papers published at Workshops, Conferences and Journals.

- ★ Francesco Azzoni, *Deep Monocular Autocalibration of Radially Symmetric Wide-Angle Cameras*.
[executive summary](#), [slides](#), published at ECCV 2024.
- ★ Andrea Marelli, *Temporal consistent CAMs for weakly supervised video segmentation in Waste Sorting*.
[executive summary](#), [slides](#), published at the Industrial Vision Workshop hosted at ECCV 2024.
- ★ Carlo Sgaravatti, *A Multimodal Hybrid Late-Cascade Fusion Network for Enhanced 3D Object Detection*.
[thesis article](#), [slides](#), published at the Industrial Vision Workshop hosted at ECCV 2024.
- ★ Olmo Michelangelo Notarianni, *Change Detection in Multivariate data streams: Online Analysis with Kernel-QuantTree*.
[executive summary](#), [slides](#), published at the Advanced Analytics and Learning on Temporal Data Workshop at ECML 2024.
- ★ Fiore Amarù, Michele Alziati, *Ensemble clustering via synchronized relabelling*.
[executive summary](#), [slides](#), published at Pattern Recognition Letters, August 2024.
- ★ Matteo Farina, *Quantum Multi-Model Fitting*.
[poster](#), published at CVPR 2023.
- ★ Andrea Porfiri Dal Cin, *Synchronization of Group-labelled Multi-graphs*.
[slides](#), published at ICCV 2022.

Other theses to have a look at:

- ★ Alessandrini, *Desc-V Desupervision in Camouflaged Videos*.
[executive summary](#), [slides](#).
- ★ Romeo, *Forcing latent space Disentanglement for enhanced model Explainability*.
[executive summary](#), [slides](#).
- ★ Rios, *Geometric Supervision for Efficient Reconstruction and Pose Refinement in Neural Radiance Fields*.
[executive summary](#), [slides](#).
- ★ Mariusz, *SE3D: A Framework for Saliency Method Evaluation in 3D Medical Imaging*.
[executive summary](#), [slides](#).

You can have a look
of some successful
thesis in the thesis
hall of fame

Computational resources



Have a look at [Server How To](#) (thanks to Luca Alessandrini)

Access to the VPN -> Machines with GPUs / a few CPUs

- Colab /Kaggle
- Data Cloud (VPN)
- Animal Machines
- HPC

General Comments

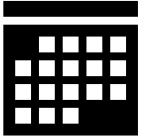
What if I disappear?



- That should not happen, this is not an option for industrial / european projects.
- We perfectly understand there can be times where you are less focused/active
- If you need to stop the thesis work for a while (e.g. exams, holidays), you have to inform us promptly.

However, if you disappear, getting back to you and start working again might be difficult. No guarantee to make it a thesis, possibly a tesina.

Thesis How To

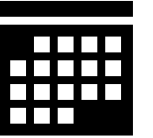


There will be a few “Thesis How To” meetings with all of you

- A. Starting your thesis work
- B. Related Work & Problem Formulation **Friday March 27th at 17.00**
- C. Problem Formulation / Solution Presentation
- D. Thesis Preparation (at least 3 months before the graduation session)
- E. Presentation Preparation (after thesis submission deadline)

Between C and D there will be primarily research activity with your co-advisor and advisor.

Do you think you are done?

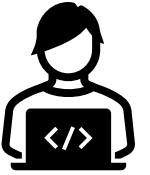


We will decide the **thesis submission time together**.

However, if you feel «ready to go» for the thesis before that, you need to inform us **at least three months** before the submission deadline...

Thesis writing requires a lot of time, we need to make sure everything we expect has been done.

Code Writing



- Use versioning
- Divide
 - training / algorithm design
 - Experiments / assessment
 - Performance evaluation and visualization
- Develop nice visualization procedures

Have a look at the «servers how-to» guide, for some guidance on training etiquette.

At the end of your work, you should **share a polished version of your code, well documented and executable.**

Your work could be the starting point for future students to further develop the research you have started.

You should agree on the way to share your work with your supervisors: private/public GitHub repository or GitLab (if you are involved in an industrial project)

Questions?